

BLOCKCHAIN BASED VOTING SYSTEM

Project Synopsis Report

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ABSTRACT

Democratic voting is a crucial and serious event in any country, the current voting scheme in any country is through ballot paper or by use of EVM. These processes have many drawbacks such as transparency, low voter turn-out, tampering of votes, distrust in the election body, forging of unique Id (voter id card), delay in giving out results and the most important is security issues. Security of digital voting is always the biggest concern when considering to implement a digital voting system. With such monumental decisions at stake, there can be no doubt about the system's ability to secure data and defend against potential attacks. One way the security issues can be potentially solved is through the use of blockchain technology. Blockchain technology offers infinite number of applications. Blockchain is a distributed ledger technology that allows digital assets to be transacted in a peer-to-peer decentralized network. A distributed ledger technology is an exciting advancement in this regard. Block is a collection of all the transactions. Blockchain possess salient features such as immutability, Decentralization, Security, Transparency and anonymity. Blockchain with smart contracts emerges as a promising candidate for building a safer, secure and transparent E-voting systems. In this paper we have implemented and tested a sample e-voting application as a smart contract for the Ethereum network using the blockchain technology through wallets and the Solidity language. Limited amount of token(gas) is given in the wallet which is exhausted when the user votes thus preventing duplicity of votes. This paper also highlights the pros and cons of using blockchain technology and also demonstrates a practical system by showcasing a webapp for voting and its limitations.

KEYWORDS: E-voting, Smart-contracts, Blockchain.

1.INTRODUCTION

Blockchain technology that shines like a star after the entrance and widespread acceptance of Bitcoin, the very first cryptocurrency in peoples' everyday life, has become a trending topic in today's software world. Blockchain technology originates from the underlying architectural design of the cryptocurrency bitcoin, where it was first introduced to the internet world and sooner became a promising technology due its high degree of transparency in the system it become an active field of research and study for its application various other fields For example, in Bitcoin, since the wallets are in a distributed structure, the total amount of coins and instant transaction volume in the world can be followed momentarily and clearly. There is no need for a central authority to approve or complete the operations on this P2P-based system.

Because of that, not only the money transfers but also all kinds of structural information can be kept in this distributed chain, and with the help of some crypto-logical methods, the system can be maintained securely. Like people's assets, marriage certificates, bank account books, medical information, etc., a lot of information can be recorded with this system with relevant modifications. Ethereum coin (Ether), another cryptocurrency with multipurpose development environments, which emerged a few years after Bitcoin, distinguishes the blockchain in a real sense, revealing that this technology can produce software that can hold information that is structured as described above. The software programs enforced by smart contracts (explained later) are written into the blockchain and are immutable. They cannot be (illegally) removed nor manipulated once written. Hence, they can work properly, autonomously and transparently forever, without any external stimuli.

Blockchain stores transactions in a block, the block eventually becomes completed as more transactions are carried out. Once complete it is then added in a linear, chronological order to the blockchain.

2.MOTIVATION

Our main motivation in this project is to provide a secure voting environment and show that a reliable e-voting scheme is possible using blockchain. Because, when e voting is available for everyone who has a computer, or a mobile phone, every single administrative decision can be made by people and members; or at least people's opinion will be more public and more accessible by politicians and managers. This will eventually lead humanity to the true direct democracy. It's important for us since elections can easily be corrupted or manipulated especially in small towns, and even in bigger cities located in corrupt countries. Plus, large-scale traditional elections are very expensive in the long term, especially if there are hundreds of geographically distributed vote centers and millions of voters. Also, the voter turn-out at the voting centers is relatively low as the person might not be staying at the address his name is enrolled in the list, or he might be out for vacation or any other work. E-voting will be able solve these problems, if implemented carefully. The concept of e voting is significantly older than blockchain. So that, all known examples so far used means of centralized computation and storage models.

3. LITERATURE REVIEW

- **Blockchain Technology Overview:**

Blockchain is a distributed ledger technology where data is stored across multiple nodes in a decentralized network. The data is immutable, transparent, and secure due to cryptographic techniques. Researchers typically begin with an exploration of how blockchain operates and why it's a promising solution for voting.

- **Challenges in Traditional Voting Systems**

Traditional voting systems (e.g., paper ballots, electronic voting machines) are prone to fraud, human error, tampering, and inefficiency. The literature often discusses these weaknesses, particularly the lack of transparency, accessibility issues, and the vulnerabilities to cyberattacks in digital voting systems.

- **Blockchain as a Solution to Voting Issues**

The central part of the literature review would focus on how blockchain addresses the identified challenges. Scholars have highlighted blockchain's core features such as immutability, transparency, and decentralization as key advantages for securing and streamlining the voting process.

- **Proposed Blockchain-Based Voting Systems**

A significant portion of the literature focuses on prototype systems and case studies where blockchain has been tested for voting. These experiments typically explore different types of blockchain (e.g., public vs. private) and smart contract functionalities for automating and verifying elections.

4. GAP ANALYSIS

A **Gap Analysis** for a **Blockchain Voting System** involves identifying the differences between the current state of research and implementation of blockchain technology in voting systems and the ideal, fully realized blockchain voting system. This analysis highlights key areas where progress is needed, and outlines challenges that have yet to be addressed in order to fully integrate blockchain into electoral processes. Here's a breakdown of the major gaps:

- Scalability and Performance.
- Voter Privacy and Anonymity.
- Public Trust and Adoption.
- Security and Resilience Against Cyber Attacks.
- Verification and Auditing.

The gap analysis reveals several critical challenges and areas where further development is needed for blockchain-based voting systems to reach their full potential. While the technology offers a promising solution to many of the weaknesses in traditional voting systems, there are significant hurdles to overcome in scalability, privacy, regulation, public trust, accessibility, and security. Bridging these gaps through research, pilot programs, and legal reforms will be crucial for the widespread adoption of blockchain in electoral systems

5. PROBLEM STATEMENT

- By adopting blockchain in the distribution of database on e-voting systems can Reduce one of the cheating sources of database manipulation.
- Blockchain technology is one of solutions, because it embraces a distributed system and the entire database are owned by many users.
- Blockchain itself has been used in the e-voting, app voting system.
- Identity Verification Challenges: Ensuring one vote per person without compromising anonymity is complex.

Problems Identified

- Election Fraud & Manipulation
- Lack of Transparency & Trust
- Security & Data Integrity Issues

Issues or Problems

- Scalability Issues
- Voter Identity & Privacy
- Security Risks

Need of Solution

- Enhanced Transparency & Trust
- Elimination of Election Fraud & Tampering
- Improved Security & Data Integrity

Existing Solutions

- Uses **biometric authentication** (facial recognition, fingerprint scanning)
- Enables remote voting via a mobile app
- Based on a **permissioned blockchain** (Hyperledger)

6. OBJECTIVES

The main objectives of a blockchain voting system are to ensure security, transparency, and trust in the voting process. By using blockchain technology, votes are securely recorded and cannot be altered, providing a tamper-proof record of all votes. This makes it much harder for anyone to manipulate the results. The system also ensures anonymity for voters, as their identities are protected while still verifying that they are eligible to vote. Additionally, blockchain's decentralized nature removes the risk of a single point of failure, like a central server being hacked. Transparency is another key objective, as everyone can verify the results independently, without relying on an external authority. Overall, the goal is to create a fair, trustworthy, and efficient voting system that guarantees the integrity of the election process

- To develop a secure and transparent voting system.
- To prevent election fraud and manipulation.
- To enable voters to verify their votes while maintaining anonymity.
- To ensure elections results are immutable and publicly auditable.

7. Tools/Technologies Used

In a blockchain voting system, several programming languages are used, depending on the specific components you're working with (**smart contracts, front-end, back-end, etc.**). Here's a breakdown of the programming languages that are commonly used:

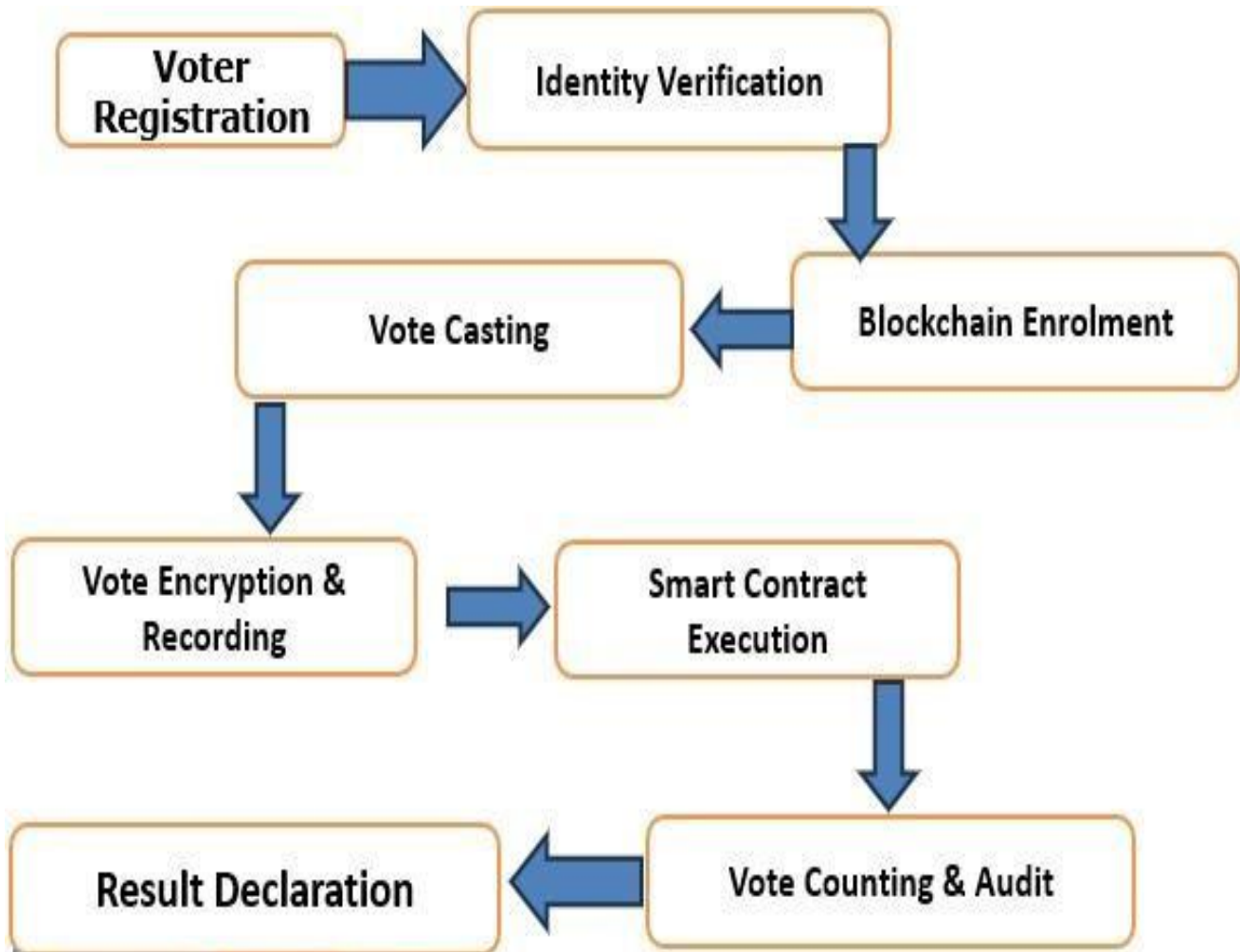
Programming languages: Solidity, JavaScript, HTML, CSS, SQL, PHP.

Summary of key programming languages used:

- **Solidity:** For smart contract development.
- **JavaScript:** For front-end development and interacting with the blockchain.
- **HTML/CSS:** For front-end structure and design.
- **SQL:** For Data storing.

8.METHODOLOGY

- **Designing the Voting Process:** First, the entire voting process is designed, which includes registering voters, casting votes, and counting them. The rules and requirements of the election (e.g., voter eligibility, vote confidentiality) are clearly defined.
- **Building the Blockchain:** A blockchain platform is chosen (like Ethereum or Hyperledger), and the structure for storing votes securely is set up. Each vote will be recorded in a block, and these blocks will be linked to form the blockchain.
- **Creating Smart Contracts:** Smart contracts are written using languages like Solidity. These are self-executing programs that automatically enforce the rules of voting. For example, smart contracts can check if a voter is eligible, count votes, and make sure that each vote is only cast once.
- **Developing the User Interface (UI):** A simple and user-friendly web or mobile application is created, where voters can log in, cast their votes, and see the results. This application interacts with the blockchain through libraries like Web3.js or Ethers.js.
- **Security Measures:** Cryptography is used to ensure that votes are secure and that voter identities are protected. Each voter's vote is encrypted, and their identity is verified without revealing who they voted for.
- **Testing and Deployment:** Once everything is set up, the system is thoroughly tested to ensure there are no bugs or security issues. After testing, the system is deployed for use in real elections.
- **Monitoring and Maintenance:** After the voting system is live, it is continuously monitored to ensure everything works smoothly, and maintenance is done to keep the system secure and up to date.



Flow chart of Blockchain based E-Voting System.

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